

DATA ANALYSIS INTERNSHIP

Day 23 REPORT

TASK - Employee Relationship Building and Dashboard Creation in Power BI

Objective

The objective of this session was to build relationships between employee datasets and create a dashboard in Microsoft Power BI using employee-related data.

Datasets Used

Two datasets were used:

1. Employee_Table

Contains:

- Employee_ID
- Employee_Name
- Department_ID
- Department_Name
- Project_ID
- Project_Name
- Salary

2. Employee_Details

Contains:

- Employee_ID
- Email
- Phone
- Attendance
- Experience_Years
- Department_ID

- Project_ID

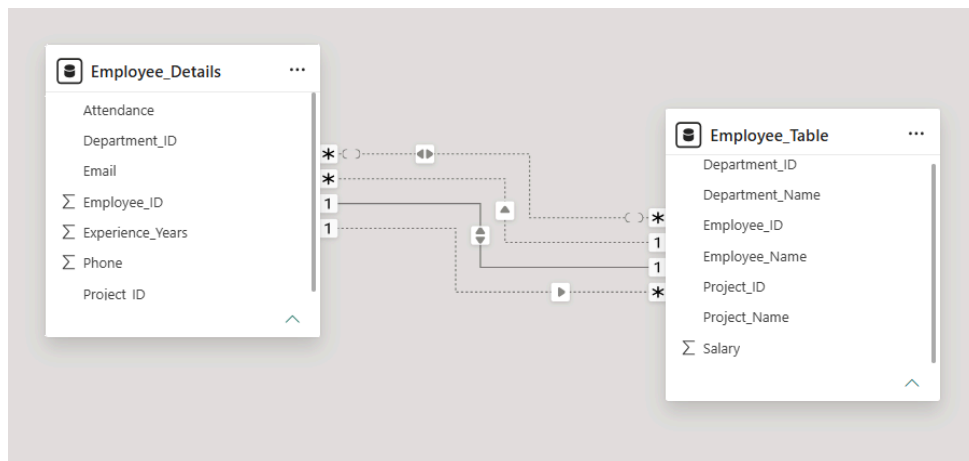
Relationship Building

Relationships were created in:

Model View

using common columns between the two tables.

The following relationship types were created:



1. One-to-Many Relationship

Connection:

Employee_Table.Employee_ID

→

Employee_Details.Employee_ID

Explanation:

One employee contains multiple detail records.

2. Many-to-One Relationship

Connection:

Employee_Details.Department_ID

→

Employee_Table.Department_ID

Explanation:

Many employees belong to one department.

3. Many-to-Many Relationship**Connection:**

Employee_Table.Project_ID



Employee_Details.Project_ID

Explanation:

Multiple employees can work on multiple projects.

Dashboard Creation

After building relationships, a dashboard was created using different visualizations for employee analysis.

Visualizations Used**1. Bar Chart****Purpose:**

Count of Project_ID by Project_Name

Projects displayed:

- AI System
 - Web App
 - Mobile App
 - Payroll System
-

2. KPI Cards

The following KPI cards were created:

KPI	Value
Sum of Salary	525K
Sum of Experience_Years	34

3. Area Chart

Purpose:

Count of Project_Name by Department_ID

Departments:

- D1
 - D2
 - D3
-

4. Pie Chart

Purpose:

Sum of Salary by Department_Name

Departments displayed:

- HR
 - IT
 - Finance
-

5. Funnel Chart

Purpose:

Count of Attendance by Department_ID

Attendance comparison was shown for:

- D1
- D2
- D3

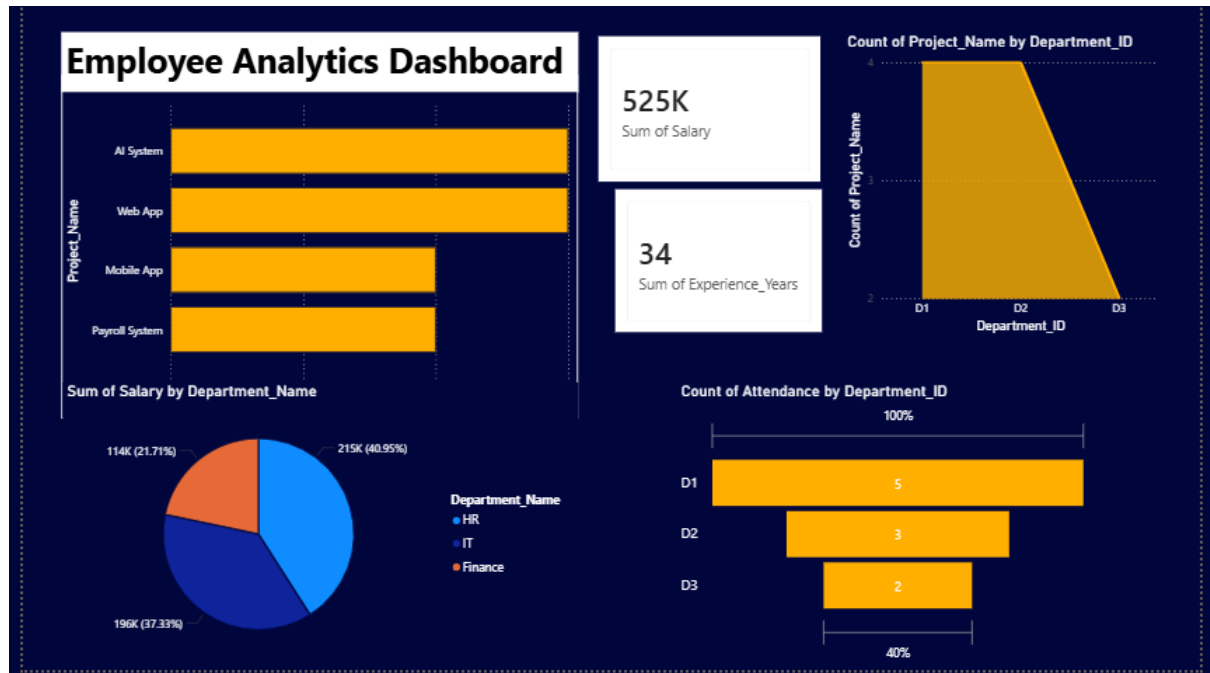
Features Used

Dark Theme Dashboard

A dark blue dashboard theme was applied for visualization.

Interactive Charts

Visuals dynamically displayed employee information and analysis.



Conclusion

In this session, relationships between Employee_Table and Employee_Details were successfully created in Power BI using Model View. Different relationship types such as One-to-Many, Many-to-One, and Many-to-Many were implemented. A dashboard was also created using bar charts, KPI cards, pie charts, area charts, and funnel charts to analyze employee data effectively.